

# OrbitWatch: Evidence-Grounded Spacecraft Telemetry Investigation

## Final-semester project technical report - generated from measured results

Experiment: nasa-full-seed17 | Completed: 2026-10-01T20:33:44.011424+00:00

Submission draft for supervisor review. This is not an approved thesis, a publication claim, or a certification of mission readiness.

### Abstract

This project implements a locally runnable spacecraft-telemetry investigation workbench. It evaluates two lightweight detectors on 81 anonymized NASA SMAP/MSL streams (506,963 evaluated samples), with chronological calibration and no point adjustment. Detected intervals become structured evidence exposed through Model Context Protocol tools. A deterministic report and a local-language-model report share a numerical validation layer. The work separates detection performance from explanation correctness, and preserves negative results and dataset limitations rather than claiming unsupported physical diagnosis.

## 1. Research question and contribution

Can evidence-constrained investigation reject unsupported numerical and causal claims while retaining useful factual coverage?

Evaluated 81 independent NASA channel files and 506,963 post-warmup test observations.

Two detectors: causal rolling-median forecasting and a small GRU; no published architecture is mislabelled.

The final 20% of each supplied training sequence calibrates residual thresholds. Test labels are evaluation-only.

MCP exposes read-only research tools. A restricted claim validator rejects unsupported values and citations.

All results are generated from saved experiment artifacts; no target accuracy is assumed.

The contribution is a reproducible systems integration and controlled explanation comparison, not a new GRU architecture or a state-of-the-art detector. Related LLM/time-series work must be discussed before making any novelty claim.

## 2. System architecture

Recorded NASA files -> channel-isolated chronological split -> forecasting -> calibrated residual threshold -> event evidence -> MCP server/client -> claim proposal -> validation -> dashboard and cited report.

The Streamlit application replays saved causal scores. Each forecast uses observations strictly before its target. The replay cursor limits visible samples, alerts, and evidence. This is not a live mission feed.

## 3. Data and provenance

NASA SMAP/MSL streams are anonymized, contain one target telemetry value plus encoded command features, and are not assumed to be synchronized across files. Published data were already scaled using test-set extrema; this upstream benchmark limitation cannot be undone. OrbitWatch fits its own scaler on training data only. Sample indices are not wall-clock timestamps. Physical root-cause labels are unavailable.

Merge duplicate annotations for the same channel by interval union, with one evaluation per unique file. The original 82-row metadata repeats SMAP P-2 with overlapping intervals; this yields 81 unique channels.

Dataset archive SHA-256: 6084d3ee3906381f2c98aa3773b6b2d77c82413503faa78f962196582e873733

Computational source SHA-256: 7397db4d000c88880302e79c0d7e22eb23f5624067bc9f63084ded7dfbfb037b

Ground-truth intervals are inclusive sample indices. Channel files are never concatenated into one timeline. The first feature is the target telemetry; remaining features are encoded commands, not a verified synchronized spacecraft sensor graph.

## 4. Detection methodology

Context window: 32 samples; GRU hidden width: 24; epochs: 4; seed: 17; maximum fitting windows/channel: 6000. Training uses Smooth L1 loss and AdamW. The standardization mean and scale are fitted only on the first training segment. The last calibration\_fraction of the supplied training file is not used for weight fitting.

Residual score  $s(t) = |\text{normalized observation}(t) - \text{normalized prediction}(t)|$ . The fixed threshold is calibration quantile 0.995. The statistical baseline predicts the median of the previous context window. The test stream has its own warmup and is not joined to the training file.

## 5. Detection results

| Detector       | Macro point F1 | Macro AP | Event recall | FP / 1000 normal |
|----------------|----------------|----------|--------------|------------------|
| gru            | 0.2162         | 0.3175   | 0.8750       | 103.13           |
| rolling_median | 0.1582         | 0.2610   | 0.6442       | 45.16            |

Macro point F1 and average precision give each channel equal weight. Event recall pools matched and true intervals. Event matching is chronological, one-to-one temporal overlap; fragmented alarms incur extra predicted events. False-positive rates accompany event metrics to discourage overly long alarms. Detection delay is in samples, not seconds. The CSV retains every channel and missed event contribution.

In this run, gru has the higher macro point F1 (0.2162). This is an observation under the declared configuration, not evidence of universal superiority. No hyperparameter selection using test labels is claimed.

## 6. Evidence-constrained explanation

Permitted claims cover the channel, event start/end, peak sample, peak ratio, observed and predicted values, and the explicit unknown-cause boundary. Each claim must cite the current evidence ID. Indices and strings match exactly; measured floating values match within 1e-6. Unsupported and duplicate claims are rejected and displayed separately.

Final prose is rendered from validated fact fields, not copied from unrestricted model text. This intentionally limits expressiveness. Correctness within this schema does not establish semantic truthfulness of arbitrary LLM explanations.

The local pilot used qwen2.5:3b. Interleaved missions, one midpoint-in-time predicted event per sorted channel; paired complete/redacted evidence. No test labels used.

Wall-clock LLM draft generation only; excludes validation and rendering. The paired validator arm reuses the ordinary draft's timing. Template generation was not timed and is reported as null, not zero.

| Mode                        | Cases | Unsupported emitted | Coverage | Mean generation seconds |
|-----------------------------|-------|---------------------|----------|-------------------------|
| grounded_local_llm          | 32    | 0.000               | 0.862    | 11.10                   |
| ordinary_llm_plus_validator | 32    | 0.000               | 0.903    | 13.29                   |
| ordinary_local_llm          | 32    | 0.118               | 0.903    | 13.29                   |
| template                    | 32    | 0.000               | 1.000    | n/a                     |

Pilot experiment, not an expert-rated semantic explanation benchmark.

The ordinary-LLM-plus-validator ablation reuses the identical ordinary-model response to isolate filtering. The grounded variant also changes the prompt.

Zero unsupported emitted facts is enforced by a restricted claim schema and trusted rendering; it is not a proof of unrestricted LLM truthfulness.

The template baseline has the same numerical correctness guarantee without an LLM.

Only numerical/identifier claims and the unknown-cause boundary are checked.

## 7. Reproducibility and implementation

README.md provides installation, training, replay, MCP, and report commands. Pinned direct dependencies and requirements-lock.txt record the software environment. The repository includes tests for temporal causality, calibration isolation, model reload, metric definitions, evidence checks, local history, actual MCP subprocess exchange, and application rendering.

MCP uses the pinned 1.x Python SDK over local stdio. The application calls tools through a real client session, rather than simulating a tool trace. All dataset artifacts, local models, and SQLite records stay outside version control.

## 8. Limitations and appropriate claims

NASA SMAP/MSL streams are anonymized, contain one target telemetry value plus encoded command features, and are not assumed to be synchronized across files. Published data were already scaled using test-set extrema; this upstream benchmark limitation cannot be undone. OrbitWatch fits its own scaler on training data only. Sample indices are not wall-clock timestamps. Physical root-cause labels are unavailable.

Single-seed measurements are not statistical significance claims.

Model configuration was specified before this evaluation; test labels do not calibrate thresholds.

All computed alarm events are included. No point adjustment or ground-truth-based alarm expansion is used.

Latency is CPU forward-pass timing, not spacecraft hardware qualification.

The explanation pilot is small and uses machine-checkable facts, not expert-labelled physical diagnoses or an operator user study. A single seed does not support confidence intervals. Anonymous benchmark preprocessing limits deployment conclusions. No hardware-energy measurement or certified onboard feasibility is claimed.

## 9. Conclusion and future work

OrbitWatch delivers an end-to-end, local, traceable research prototype and a reproducible comparison. Its primary value is the separation of numerical evidence from unsupported interpretation. Extensions should prioritize repeated-seed evaluation, independent expert annotation, operationally meaningful datasets, and calibration robustness rather than adding unvalidated model names.

## References

Hundman et al. Detecting Spacecraft Anomalies Using LSTMs and Nonparametric Dynamic Thresholding. KDD 2018. <https://arxiv.org/abs/1802.04431>

Kim et al. Towards a Rigorous Evaluation of Time-series Anomaly Detection. <https://arxiv.org/abs/2109.05257>

Alnegheimish et al. Can Large Language Models be Anomaly Detectors for Time Series? DSAA 2024. <https://arxiv.org/abs/2405.14755>

NASA benchmark implementation and data documentation: <https://github.com/khundman/telemanom>

Model Context Protocol Python SDK: <https://github.com/modelcontextprotocol/python-sdk>